

Portfolio of Burcu Erarslanoglu

Signal Processing • Machine Learning • Medical Imaging

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This portfolio provides quick access to my recent engineering work. All blue text throughout this document is hyperlinked to GitHub where associated reports and codes can be found.

Deep Learning Guided Reconstruction for Cardiac Cine MRI

Nov 2024 - Jul 2025

- Demonstrated that domain-mismatched data with temporal structure can serve as effective training data for cardiac cine MR reconstruction.
- Found that temporal convolutions in U-Net architecture improve reconstruction quality by exploiting shared information across adjacent cardiac frames.
- Showed that temporal kernel size has a non-trivial effect: moderate context outperforms larger context under high accelerations.
- Demonstrated that time-varying k-space sampling further improves reconstruction when combined with temporal convolutions.

Gaussian Process Regression on a Grid with Tensor Train Matrices

Sep 2024 - Nov 2024

- Implemented Gaussian Process Regression for large grid-structured datasets using tensor train representations to exploit Kronecker structure in covariance matrices.
- Demonstrated significant reduction in computational complexity while preserving prediction accuracy.

Hand Gesture Classification Based on EMG Signals

Sep 2024 - Nov 2024

- Developed a machine learning pipeline for hand gesture classification from EMG signals collected across multiple subjects.
- Extracted features in both time and frequency domains and evaluated KNN, SVM, and MLP classifiers using cross-validation.
- Optimized model performance through hyperparameter tuning and comparative analysis across classifiers.

A Review of Cardiac MRI Reconstruction Techniques

Sep 2024 - Nov 2024

- Reviewed traditional and emerging cardiac MRI reconstruction techniques, including parallel imaging, compressed sensing, deep learning, and hybrid approaches.
- Conducted a comparative analysis of these methods and assessed their strengths, limitations, and applicability to cardiac cine MRI.
- This laid the foundation for the background and literature review sections of my Master's thesis, *Deep Learning Guided Reconstruction for Cardiac Cine MRI*.

Deriving FC with fMRI Using BSS with Tensor Decomposition

Apr 2024 - Aug 2024

- Reviewed blind source separation and tensor decomposition methods for extracting functional connectivity from fMRI data.
- Conducted comparative analyses of these methods and assessed their strengths and limitations for high-dimensional neuroimaging applications.

Linear Discriminant Analysis for Recognising Differences

Apr 2024 - Jun 2024

- Applied a PCA-LDA framework to MRI brain scans to classify Alzheimer's disease through dimensionality reduction.
- Evaluated classification accuracy across different numbers of principal components using k-fold cross-validation, demonstrating potential for early diagnosis of neurodegenerative diseases.

Array Parameter Estimation and Speech Enhancement

Apr 2024 - Jun 2024

- Studied methods used in the estimation of directions and frequencies in an array and channel equalization.
- Compared the performances of EVD and GEVD on the three different beamformers by using SNR and STOI metrics.

Wavefield Imaging on 2D Linearized Inverse Scattering Problems

Apr 2024 - Jun 2024

- Solved a 2D linearized inverse scattering problem to reconstruct an image of an object from wavefield measurements using contrast function estimation.

Boolean Compressed Sensing

Feb 2024 - May 2024

- Implemented and compared several sparse recovery algorithms for Boolean compressed sensing in the context of group testing.

Compressed Sensing and Machine Learning Techniques in MRI

Feb 2024 - May 2024

- Reviewed compressed sensing and machine learning approaches for accelerated MRI, covering techniques and applications across both domains.
- Implemented a compressed sensing + DnCNN pipeline in MATLAB for brain MRI reconstruction.
- Showed that ML-assisted undersampling can recover SNR comparable to fully sampled acquisitions while reducing scan time.

Towards Reconstructing the Brain with Graph Signal Processing

Feb 2024 - Apr 2024

- Conducted a literature review on graph signal processing (GSP) for modelling structural and functional brain connectivity.
- Surveyed approaches combining GSP with deep learning for neuroimaging analysis and connectivity matrix generation, and assessed their potential for mental disorder diagnosis.

Compressed Sensing for Applied Convex Optimisation

Nov 2023 - Jan 2024

- Formulated signal reconstruction as a convex optimization problem and implemented a compressed sensing-based recovery algorithm.
- Reconstructed signals using both off-the-shelf solvers and a custom low-complexity projected subgradient descent algorithm, and compared their performance.

Accelerometer Calibration for Estimation and Detection

Nov 2023 - Jan 2024

- Applied Maximum Likelihood Estimation (MLE) to calibrate accelerometer measurements and derived the Cramér-Rao Lower Bound (CRLB) to benchmark estimator performance.

Fusing Noisy and Corrupted MRI Images

Sep 2023 - Nov 2023

- Developed a pipeline to detect and restore outliers in eye MRI scans caused by patient motion and noise, using optimal filtering and spectral estimation techniques.
- Implemented and compared multiple filters, evaluating image quality improvements quantitatively using SNR.

Detection of Forest Fires with WSNs

Sep 2022 - July 2023

- Proposed and tested wireless sensor network (WSN) algorithms for forest-fire detection via MATLAB simulation, as part of a capstone project with METEKSAN Defence.
- Evaluated sensor placement, power efficiency, communication protocols, and terrain-propagation effects to optimize detection performance.

Online Music Platform Database Application

Feb 2023 - June 2023

- Designed and implemented a database-backed online music platform supporting multiple user roles.
- The system enabled operations such as track browsing/filtering, album/playlist management, user history tracking, and social interactions, including following users and liking playlists.
- The project involved designing relational database schemas and implementing queries for managing music metadata, user activity, and artist collaborations.

Medical Imaging Project: Brain MRI Simulator

Dec 2022 - Jan 2023

- Built an MRI simulator from scratch to cover the full acquisition pipeline: slice selection, k-space sampling, and image reconstruction.
- Studied the main contrast mechanisms of MRI, namely T_1 , T_2 , PD weighting, and flip angle, alongside

Gaussian and spike noise effects.

- Investigated the effect of each contrast mechanism and noise type on simulated brain MR images.

CMOS VLSI Design Project: SRAM Design

May 2022 - Jun 2022

- Designed and combined an 8x4 memory array and 8x4 SRAM array in Cadence Virtuoso, validating functionality via testbench simulation and optimizing for power.

Electronic Circuit Project: Design of D and T Flip-Flop

Sep 2021 - Dec 2021

- Designed D and T flip-flops using BJTs; simulated in LTspice, validated on a breadboard, and implemented on a custom PCB designed in DipTrace.

Microprocessors Project: Smart Car

Feb 2021 - May 2021

- Built an autonomous smart car on the FRDM-KL25Z board (C++, Mbed) that used color and ultrasonic sensing for line-following and obstacle avoidance, with keypad input and dot-matrix display feedback.

Digital Circuit Design Project: Weather Station

Sep 2020 - Dec 2020

- Designed a digital weather station on the BASYS3 FPGA board (VHDL), interfacing DHT22 and LPS25HB sensors with LCD and VGA display outputs for real-time environmental readouts.